20

Computational Constraints

▶ 20.1 Inference Interpretations of Algorithms vs Algorithms for Inference

In this chapter, we have so far established that there are self-contained probabilistic inference algorithms on matrix elements whose behaviour is consistent with that of existing linear solvers. These corroborate the view that computation is inference.

On a more practical level, a corollary of these results is that we can use *existing* – efficient, stable – implementations of linear solvers, CG in particular, and treat them as a source of data, producing informative action–output pairs $(s_i, y_i = As_i)_{i=1,\dots,M}$. Combined with the kind of structured Gaussian priors studied above, these data then give rise to posteriors on A and H, and these posteriors have convenient properties: their posterior mean is a sum of A_0 and a term of low rank, thus easy to handle both analytically and computationally. And the good convergence properties[1] of CG translate into desirable convergence properties of the Gaussian posterior.

Thus, consider the data set $S, Y \in \mathbb{R}^{N\times M}$ collected by running CG on A, b. If we adopt this practically minded approach, the desiderata for the Gaussian prior change. It no longer matters whether the prior is particularly consistent with the actions of the algorithm that collected the data. Instead, two new considerations arise:

Computational efficiency: The Gaussian posterior should have particularly low storage and evaluation cost. In particular, one would like to use the known properties of CG – orthogonal gradients, conjugate directions.

Uncertainty calibration: The posterior covariance – the main

[1] Expositions on the convergence of CG and related methods can for example be found in §5.1 (p. 112 onwards) of Nocedal and Wright (1999) and in §11.4.4. of Golub and Van Loan (1996). A frequently used result is that CG after $k+1$ steps, assuming exact computations, finds the solution $x_{k+1} = P_k^*(A)r_0$, where P_k^* is the (matrix) polynomial of degree k that solves the following optimisation problem over all such polynomials:

$$P_k^* = \arg\min_{P_k} \|x_0 + P_k(A)r_0 - x\|_A, \tag{20.1}$$

where x is the true solution of $Ax = b$, and $\|v\|_A^2 := v^\intercal Av$. This result can be used to phrase the convergence in terms of the eigenvalue spectrum of A (e.g. p. 116 in Nocedal and Wright (1999)). In particular, if A has eigenvalues $\lambda_1 \leq \dots \leq \lambda_N$, then the error of CG after $M+1$ steps is roughly given by

$$\|x_{M+1} - x\|_A \approx (\lambda_{N-M} - \lambda_1)\|x_0 - x^*\|_A. \tag{20.2}$$

Simply put, if A has $K \ll N$ large eigenvalues and $N-K$ small ones, then CG finds a good estimate in only K steps. Since these optimisation steps are taken in the span of S_M (the Krylov sequence of Eq. (18.2)), this also means that the low-rank term in the posterior mean A_M approximately covers the dominant eigenvalues of A.

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"added value" of the probabilistic solver over a classic solver – should be analytically linked to the estimation error. We will find below that this intuitive desideratum is not straightforward to translate into a concrete analytical statement. Since the object of interest is matrix-valued, design criteria focussing on different aspects of the matrix lead to different choices for the prior.

Assume we want to perform inference on A with a symmetry-encoding prior $p(A) = \mathcal{N}(A; A_0, W_0 \otimes W_0)$. Both the computational and the calibration viewpoints suggest choosing

$$\begin{aligned} &A_0 = 0, \text{ and} \\ &W_0 \text{ such that } W_0 S = Y. \end{aligned}$$

Computationally, this choice is appealing, because then the Gram matrix $S^\intercal W_0 S = S^\intercal Y = S^\intercal AS$ is diagonal (since CG produces conjugate directions), and to set $A_0 = 0$, and all terms with $S^\intercal A_0 S$ in the posterior vanish.[2] From Eq. (19.21), the posterior mean is then given simply by

$$A_M = Y(S^\intercal Y)^{-1} Y^\intercal = \sum_{m=1}^{M} \frac{y_m y_m^\intercal}{s_m^\intercal y_m} = \tilde{Y}\tilde{Y}^\intercal, \qquad (20.3)$$

and can thus be stored in form of the $N \times M$ matrix $\tilde{Y} = Y(S^\intercal Y)^{-1/2}$. Since for SPD A, $S^\intercal Y = S^\intercal AS$, is a diagonal matrix with only positive entries on the diagonal, its matrix square root can be computed simply from the real numbers $\sqrt{s_m^\intercal y_m} > 0$.

[2] A scalar $A_0 = \alpha_0 I$ is also relatively easy to handle. Because CG produces orthogonal gradients and $y_i = As_i = r_i - r_{i-1}$, the matrix $Y^\intercal Y$ is symmetric tridiagonal positive definite:

$$\begin{aligned} (Y^\intercal Y)_{ij} &= y_i^\intercal y_j = (r_i - r_{i-1})^\intercal (r_j - r_{j-1}) \\ &= \delta_{ij}(r_i^\intercal r_i + r_{i-1}^\intercal r_{i-1}) \\ &\quad + \delta_{i,j-1} r_{i-1}^\intercal r_{i-1} \\ &\quad + \delta_{i,j+1} r_i^\intercal r_i. \end{aligned}$$

Tridiagonal problems can be efficiently solved in $\mathcal{O}(M)$ time and thus can be treated as "essentially trivial". For SPD tridiagonal systems, a concrete example is Algorithm 4.3.6 (p. 181) of Golub and Van Loan (1996). This algorithm requires $8M$ floating point operations to solve the $M \times M$ system). In LAPACK (Anderson et al., 1999), the corresponding solvers are called `xPTSV` (for Positive definite Tridiagonal SolVer, with `x=S,D,C,Z` for single, double, complex, double complex, respectively).

Let us again consider the hypothetical choice

$$W_0 = A.$$

We find that it is also favourable from the viewpoint of uncertainty calibration: the prior $p(A) = \mathcal{N}(A; 0, W_0 \otimes W_0)$ assigns element-wise variance

$$\operatorname{var}_{p(A)}([A]_{ij}) = \frac{1}{2}([W_0]_{ii}[W_0]_{jj} + [W_0]_{ij}^2). \qquad (20.4)$$

Thus, for elements of the diagonal, we have what might be called *perfect* calibration – the true square error equals the expected square error:

$$\frac{\left([A]_{ii} - \mathbb{E}_{p(A)}([A]_{ii})\right)^2}{\mathbb{E}_{p(A)}\left(\left([A]_{ii} - \mathbb{E}_{p(A)}([A]_{ii})\right)^2\right)} = 1. \qquad (20.5)$$

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For off-diagonal elements, the variance is an upper error bound

$$\frac{\left([A]_{ij} - \mathbb{E}_{p(A)}([A]_{ij})\right)^2}{\mathbb{E}_{p(A)}\left(\left([A]_{ij} - \mathbb{E}_{p(A)}([A]_{ij})\right)^2\right)} = \frac{[A]_{ij}^2}{\frac{1}{2}([A]_{ii}[A]_{jj} + [A]_{ij}^2)} < 1, \tag{20.6}$$

because we assumed A to be SPD, and thus

$$|[A]_{ij}| \leq \sqrt{[A]_{ii}[A]_{jj}}, \qquad \forall\, 0 < i, j \leq N.$$

It would of course be preferable to achieve perfect calibration for *all* matrix elements, not just on the diagonal. But the symmetric Kronecker structure imposes restrictions on the form of the covariance, so we have to live with some trade-off. This situation is analogous to that we encountered in the chapter on integration: to infer integrals, we had to model the integrand with a GP whose covariance function (kernel) was chosen such that the posterior mean be analytically *integrated*. This cannot be achieved in general with perfect calibration, so we settled for under-confidence. In this chapter, to infer matrix inverses, we have to model the matrix with a Gaussian whose covariance is chosen such that the posterior mean can be analytically *inverted*. Again, we find that perfect calibration cannot be achieved, and settle for under-confidence.

Corollary 19.9 further supports the choice

$$W_0 = A, \qquad A_0 = 0,$$

since A_M is then always positive semi-definite. Setting $A_0 = 0$ yields a rank-deficient estimator A_M, so we cannot use the matrix inversion lemma to compute an inverse. However, the pseudoinverse[3] of A_M can be computed efficiently and has the right conceptual properties for many applications. For a factorised symmetric matrix like our $A_M = \tilde{Y}\tilde{Y}^\intercal$, the pseudoinverse is given by

$$A^+ = \tilde{Y}(\tilde{Y}^\intercal\tilde{Y})^{-2}\tilde{Y}^\intercal.$$

Since $\tilde{Y}^\intercal\tilde{Y}$ is tridiagonal symmetric positive definite, its inverse can be computed in $8M$ operations (see note 2).

Alas, we can of course not set $W_0 = A$, since A is the very matrix we are trying to infer. We could set

$$W_0 = A_M$$

in what might be called an empirical Bayesian approach. This would ensure $WS = Y$. But then the posterior variance vanishes

[3] The Moore–Penrose pseudoinverse A^+ of a matrix A is a generalisation of the inverse A^{-1} that exists for any matrix A. It is defined (for real-valued matrices) as a matrix with the properties

$$\begin{aligned} AA^+A &= A, \\ A^+AA^+ &= A^+, \\ (AA^+)^\intercal &= AA^+, \\ (A^+A)^\intercal &= A^+A. \end{aligned}$$

(The concept seems to have been invented by Fredholm (1903) for operators, and discussed for matrices by Moore (1920).) The pseudoinverse yields the least-squares solution A^+b for our linear problem $Ax = b$ in the sense that

$$\|Ax - b\|_2 \geq \|AA^+b - b\|_2 \quad \forall x \in \mathbb{R}^N.$$

For the choice $A_0 = 0$, A_M^+ can also be seen as the natural limit of the estimator A_M^{-1} arising from $A_0 = \alpha I$ for small α, because, for general A,

$$\begin{aligned} A^+ &= \lim_{\alpha \to 0} (A^\intercal A + \alpha I)^{-1} A^\intercal \\ &= \lim_{\alpha \to 0} A^\intercal (AA^\intercal + \alpha I)^{-1}. \end{aligned}$$

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(from Table 19.1):

$$\begin{aligned} W_M &= W_0 - W_0 S(S^\intercal W_0 S)^{-1} S^\intercal W_0 \\ &= \tilde{Y}\tilde{Y}^\intercal - \tilde{Y}\tilde{Y}^\intercal = 0. \end{aligned}$$

So we will have to do something more elaborate. Section 21 discusses ways to estimate a prior covariance parameter W_0 that is consistent with the choice of A_M and achieves non-zero posterior variance according to different desiderata for calibration.

▶ 20.2 Inferring the Solution x

Across this chapter, the linear problem $Ax = b$ has been phrased in terms of finding the matrix inverse A^{-1}. This provides a way to find x for *any* $b \in \mathbb{R}^{N \times N}$. Depending on the application, however, it may be entirely enough to just solve for one specific b. In such cases the detour through the matrix inverse is not necessary, and we may wonder whether we can get away with less computational overhead. A derivation and analysis of linear solvers phrased as inference on x can be found in Cockayne et al. (2019a); the connection to matrix-based inference is further explored in Bartels et al. (2019).

This setting is where a formulation with a prior on H is more convenient than one with a prior on A. Because $x = Hb$ is a linear map of H, any Gaussian prior

$$p(H) = \mathcal{N}(H; H_0, \Sigma) \qquad \text{with } \Sigma \in \mathbb{R}^{N^2 \times N^2}$$

directly translates into a Gaussian prior on x, with

$$\begin{aligned} p(x = Hb) &= \mathcal{N}(x; H_0 b, (I \otimes b)^\intercal \Sigma (I \otimes b)) \\ &=: \mathcal{N}(x; x_0, \Xi_0) \qquad \text{with } \Xi_0 \in \mathbb{R}^{N \times N}. \end{aligned}$$

The (noise-free) observations $AS = Y$ can then be interpreted in two different ways: either as linear projections of H of the form $S = HY$, or as linear projections of $x = H^\intercal b$ of the form

$$Y^\intercal x = S^\intercal b. \tag{20.7}$$

Equation (20.7) is a statement about only the b numbers in $Y^\intercal H^\intercal b = S^\intercal b \in \mathbb{R}^M$ rather then the $N \times M$ numbers in S, highlighting that inference on x is more limited, but also less expensive, than explicit inference on H followed by a projection on b. The direct Gaussian posterior on x has the form

$$p(x \mid Y^\intercal x = S^\intercal b) = \mathcal{N}(x; x_M, \Xi_M), \tag{20.8}$$

$$\text{with} \quad x_M = x_0 + \Xi_0 Y (Y^\intercal \Xi_0 Y)^{-1} (S^\intercal b - Y^\intercal x_0),$$

$$\text{and} \quad \Xi_M = \Xi_0 - \Xi_0 Y (Y^\intercal \Xi_0 Y)^{-1} Y^\intercal \Xi_0. \tag{20.9}$$

For example, the non-symmetric prior $p(H) = \mathcal{N}(H; H_0, V \otimes W)$ induces the prior $p(x) = \mathcal{N}(x; x_0, \Xi_0)$ with $x_0 = H_0^\intercal b$ and

$$\Xi_0 = (I \otimes b)^\intercal (W_0 \otimes V_0)(I \otimes b) = (b^\intercal V_0 b) W_0.$$

On the other hand, the symmetric Kronecker-structured prior

$$\begin{aligned}
p(H) &= \mathcal{N}(H; H_0, W_0 \circledast W_0) \text{ induces} \\
p(x) &= \mathcal{N}(x; x_0, \Xi_0) \text{ with } x_0 = H_0 b \text{ and} \\
\Xi_0 &= (I \otimes b)^\intercal (W_0 \circledast W_0)(I \otimes b) \\
&= \frac{1}{2}\left(W_0(b^\intercal W_0 b) + W_0 b b^\intercal W_0\right) \\
&=: \frac{1}{2}\left(\beta W_0 + \tilde{b}\tilde{b}^\intercal\right) \quad \text{with } \tilde{b} := W_0 b, \beta := b^\intercal W_0 b.
\end{aligned}$$

The matrix inversion lemma yields

$$\begin{aligned}
(Y^\intercal \Xi_0 Y)^{-1} &= \frac{1}{2}\left(\beta (Y^\intercal W_0 Y) + Y^\intercal \tilde{b}\tilde{b}^\intercal Y\right)^{-1} \\
&= 2\beta^{-1}(Y^\intercal W_0 Y)^{-1}\left(I - \frac{Y^\intercal \tilde{b}\tilde{b}^\intercal Y (Y^\intercal W_0 Y)^{-1}}{\beta + \tilde{b}^\intercal Y (Y^\intercal W_0 Y)^{-1} Y \tilde{b}}\right),
\end{aligned}$$

and we see from Eqs. (20.8)–(20.9) that conditioning on $S = HY$ gives rise to a posterior on x with mean

$$x_M = x_0 + (\beta W_0 Y + \tilde{b}\tilde{b}^\intercal Y)\beta^{-1}(Y^\intercal W_0 Y)^{-1}\left(I - \frac{Y^\intercal \tilde{b}\tilde{b}^\intercal Y (Y^\intercal W_0 Y)^{-1}}{\beta + \tilde{b}^\intercal Y (Y^\intercal W_0 Y)^{-1} Y \tilde{b}}\right)(S^\intercal b + Y^\intercal x_0). \quad (20.10)$$

While these terms have a tedious structure, they can be managed efficiently by keeping track of the M-dimensional vector $b^\intercal W_0 Y$ and the $M \times M$ matrix $Y^\intercal W_0 Y$. As before, this can be achieved in a particularly efficient manner if the choices W_0 and Y well-chosen relative to each other: if the observations Y are constructed by running CG and $W_0 = \omega I$, then $Y^\intercal Y$ is a tridiagonal SPD matrix.

Exercise 20.1 (medium). *The more general yet more expensive path is to condition the prior $p(H) = \mathcal{N}(H; H_0, W_0 \circledast W_0)$ on the $N \times M$ observations $S = HY$, then project onto x. The resulting posterior mean is $x_M = H_M b$, with H_M from Table 19.1. Compute this projection $H_M b$ and convince yourself that it has a more complicated form than Eq. (20.10) (e.g. it explicitly contains terms in H_0). What is the significance of these additional terms? Do they change the value of x_M? (Hint: Consider their effect on the space outside the span of $Y^\intercal W_0 b$.)*

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